



CSE 251

Electronic Devices and Circuits

Lecture 7

Course instructor:

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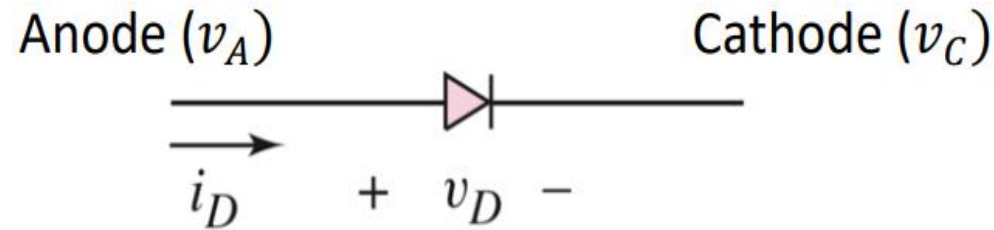
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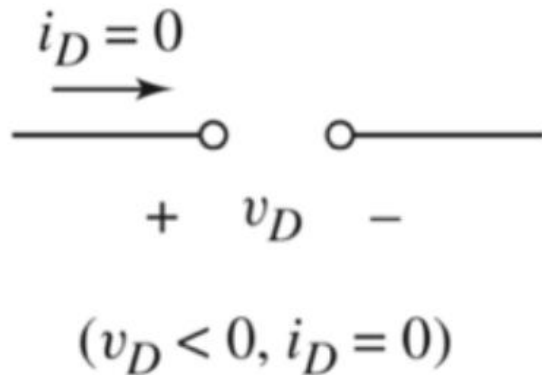
Outline

- **Diode Models and IV Characteristics: Review**
- **Diode Logic OR operation**
- **Diode Logic AND operation**

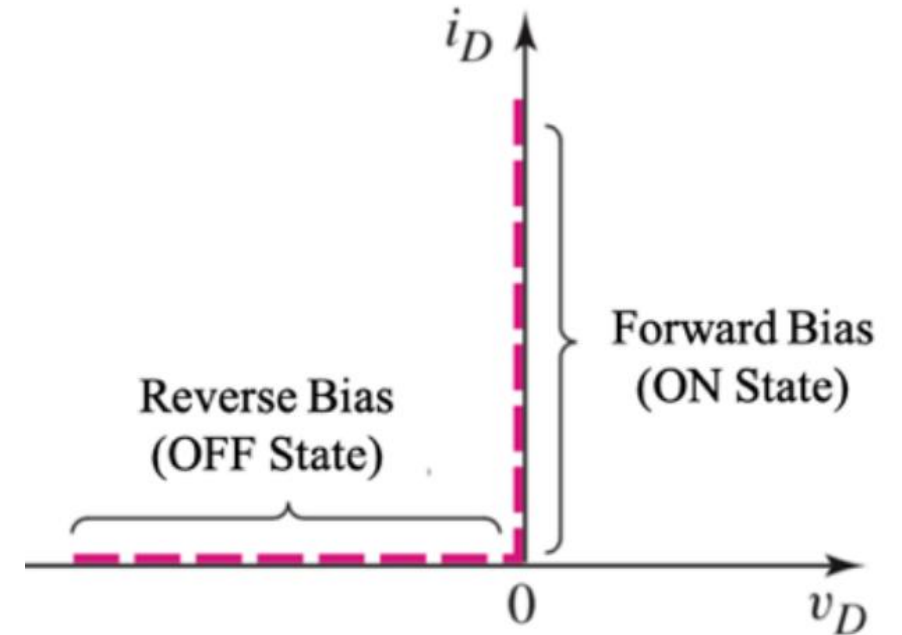
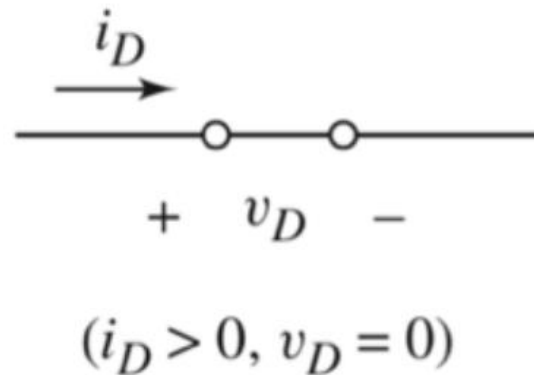
Review: Ideal Diode Model



OFF State: Open circuit

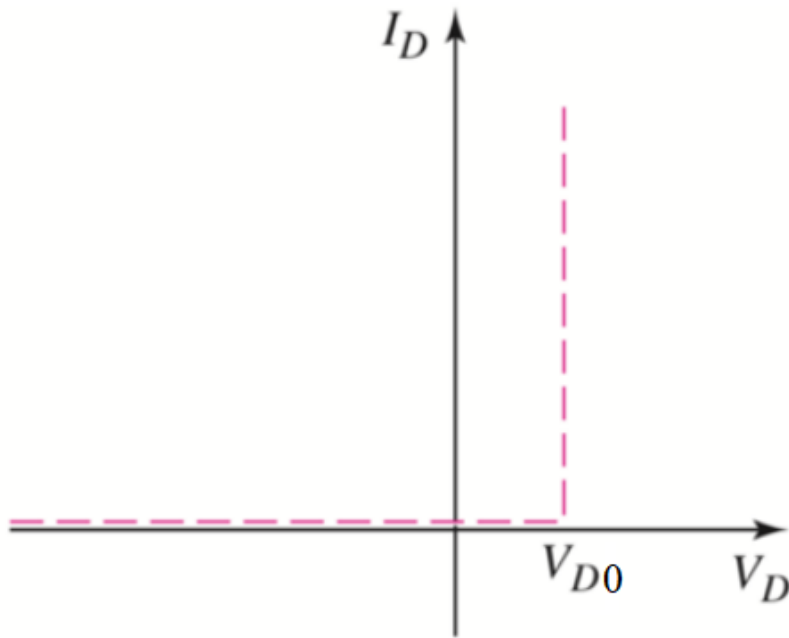


ON State: Short circuit

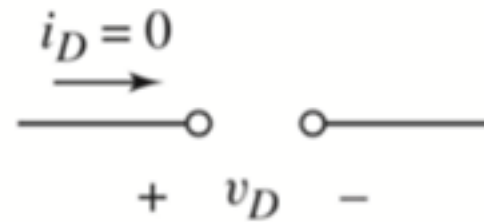


Modeling the real diode

1. Ideal diode model
- 2. Constant voltage drop (CVD) model**
3. CVD+R model

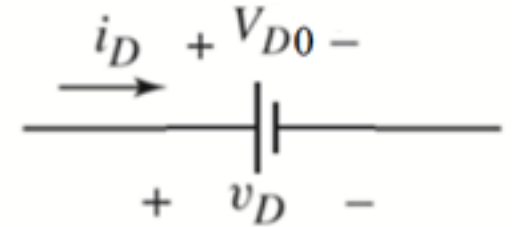


OFF State: Open circuit



$$(v_D < V_{D0}, i_D = 0)$$

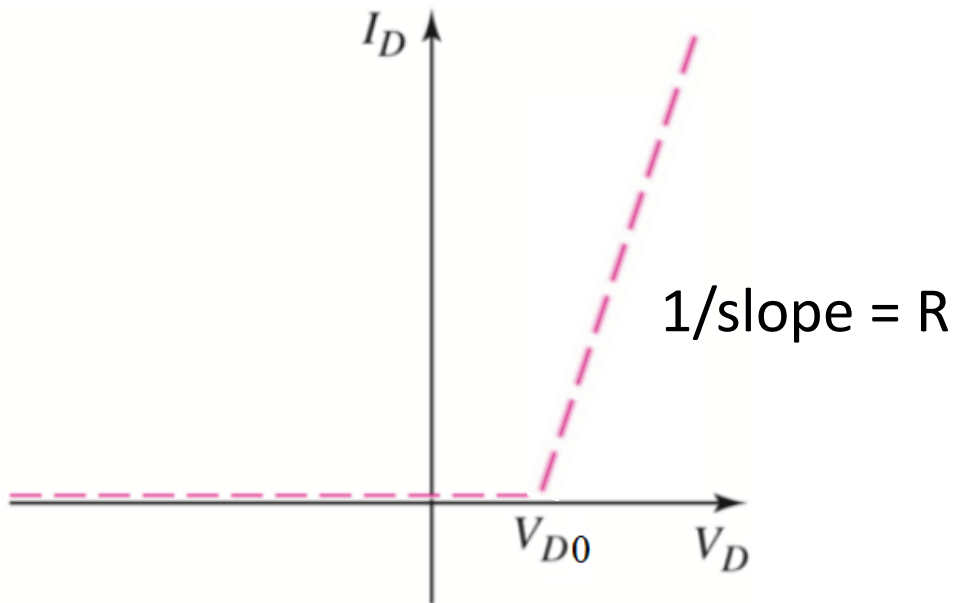
ON State: Voltage source



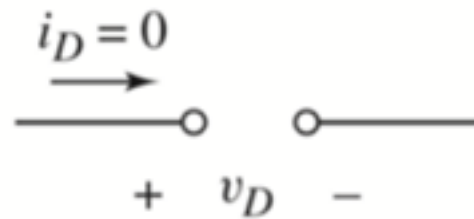
$$(i_D > 0, v_D = V_{D0})$$

Modeling the real diode

1. Ideal diode model
2. Constant voltage drop (CVD) model
- 3. CVD+R model**

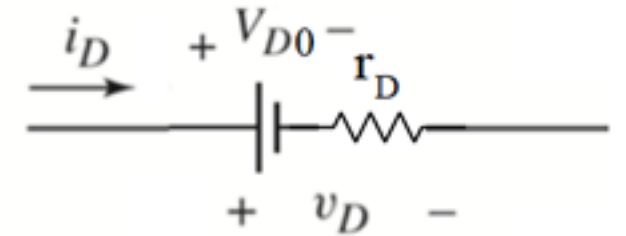


OFF State: Open circuit



$$(v_D < V_{D0}, i_D = 0)$$

ON State: Voltage source



$$(i_D > 0, v_D = V_{D0} + i_D r_D)$$

Digital Representation

- Binary $\rightarrow$ Two states (0/False, 1/True)
- Binary variables in circuit, need to use two states of device/parameters

Voltage

5V $\rightarrow$ 1

0V $\rightarrow$ 0

Current

2mA $\rightarrow$ 1

3mA $\rightarrow$ 0

State

Diode

ON $\rightarrow$ 1

OFF $\rightarrow$ 0

Memristor

Low resistance $\rightarrow$ 1

High resistance $\rightarrow$ 0

Logical Operations with Diode (OR)

Logic Truth Table

INPUTS		OUTPUT
X	Y	Z
0	0	0
0	1	1
1	0	1
1	1	1

Voltage Truth Table

INPUTS		OUTPUT
X	Y	Z
0 V	0 V	0 V
0 V	5 V	5 V
5 V	0 V	5 V
5 V	5 V	5 V

Logic Levels:

Corresponding voltage levels:

Low/False

0

0V

High/True

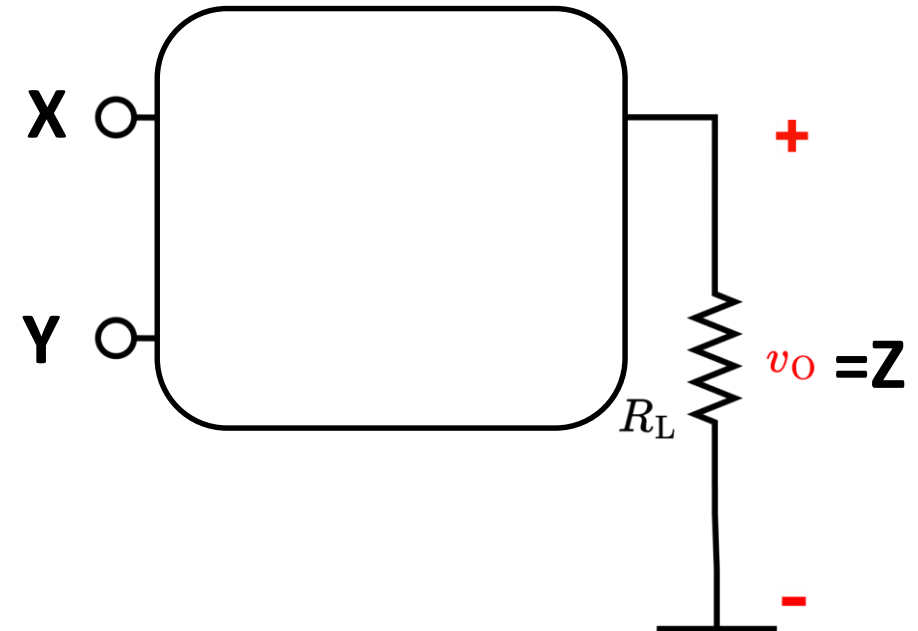
1

5V

Logical Operations with Diode (OR)

Voltage Truth Table

INPUTS		OUTPUT
X	Y	Z
0 V	0 V	0 V
0 V	5 V	5 V
5 V	0 V	5 V
5 V	5 V	5 V



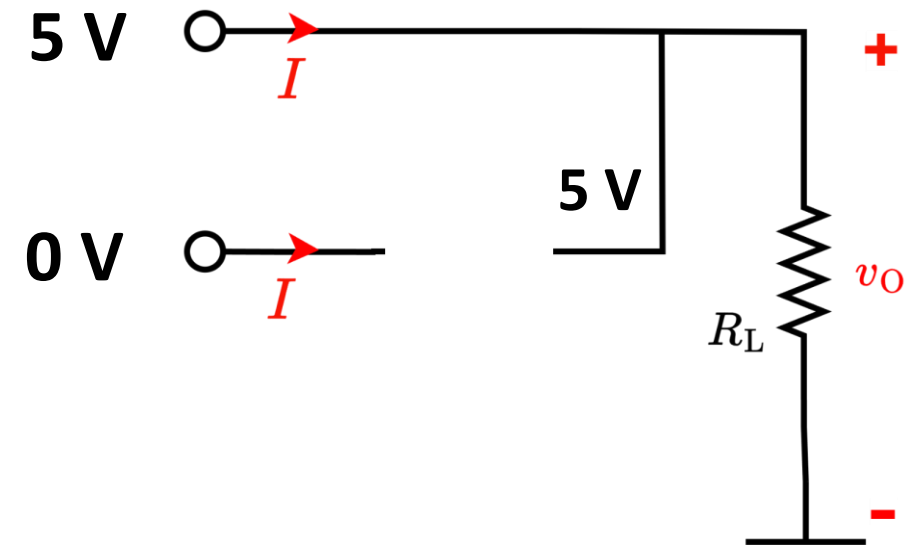
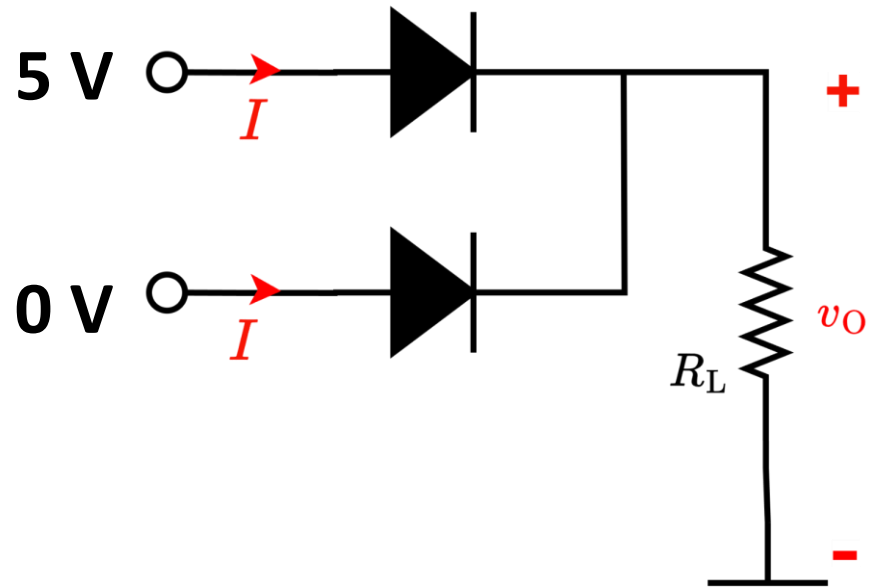
PULL DOWN NETWORK

When all inputs are completely disconnected, v_O is pulled down to **GND**

Degrades the HIGHEST output voltage

Logical Operations with Diode (OR)

Ideal diode

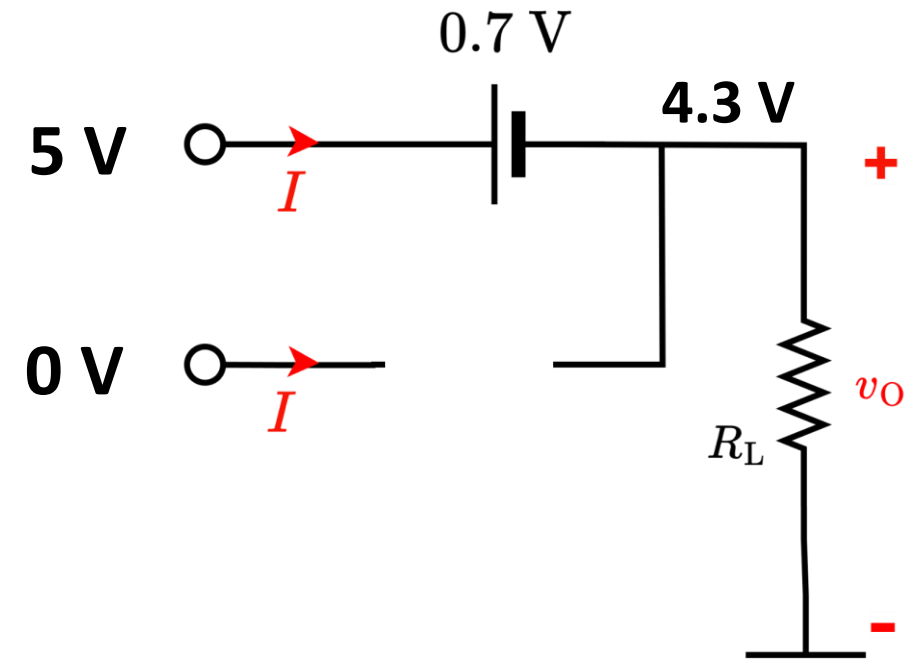
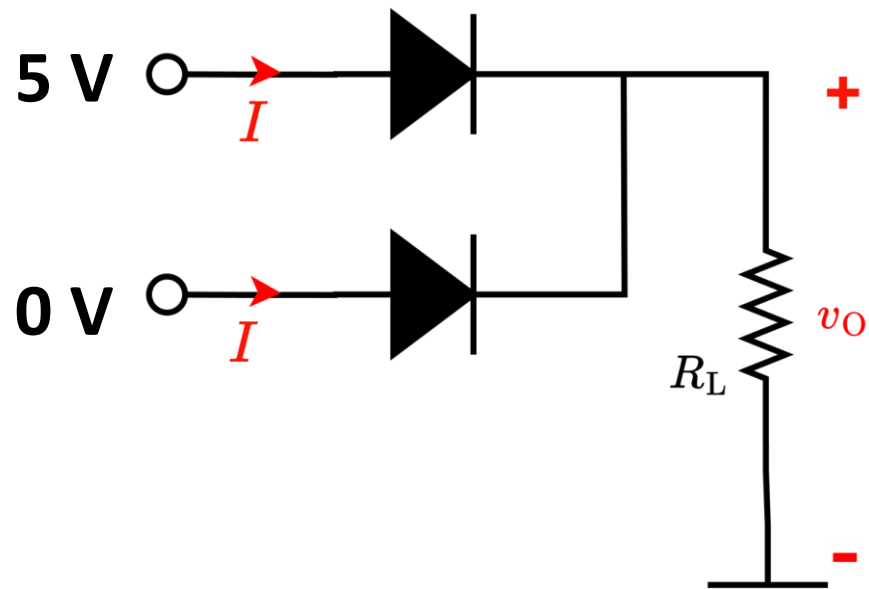


$$Z = 5\text{ V}$$



Logical Operations with Diode (OR)

CVD diode



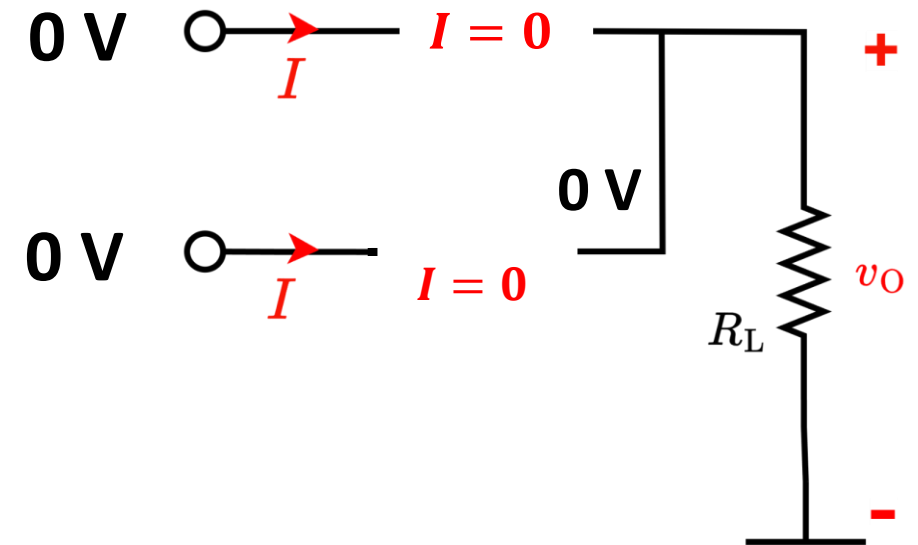
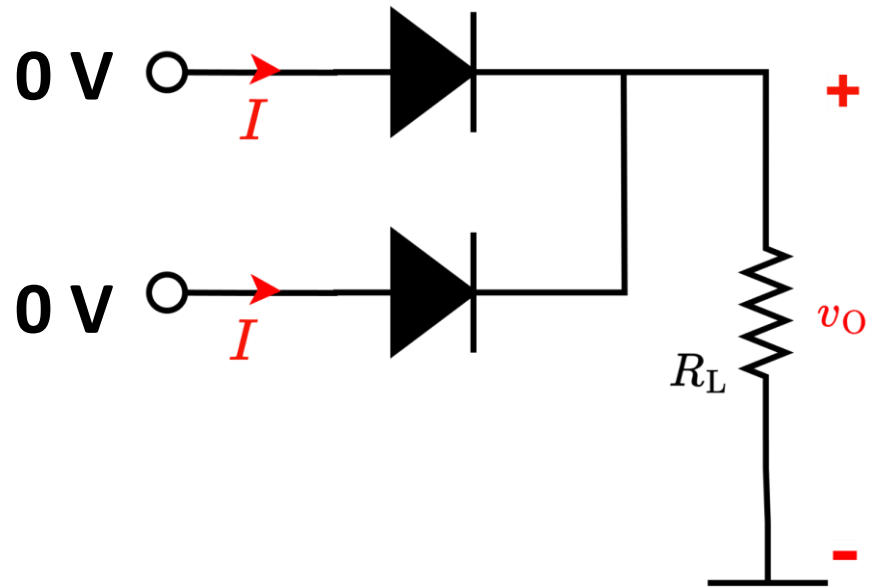
$$Z = 4.3\text{ V}$$

Degraded 5 V



Logical Operations with Diode (OR)

Ideal diode

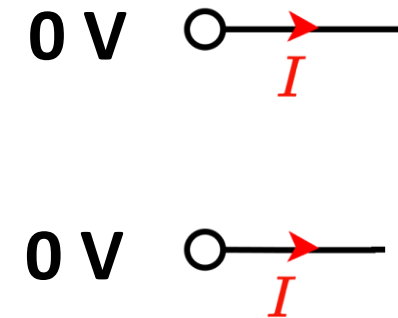
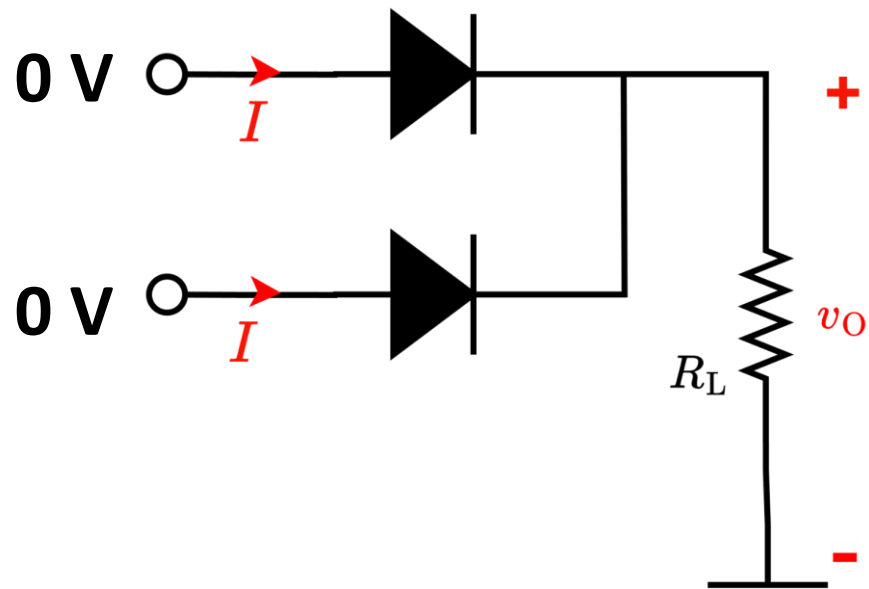


$$Z = 0\text{ V}$$



Logical Operations with Diode (OR)

CVD diode

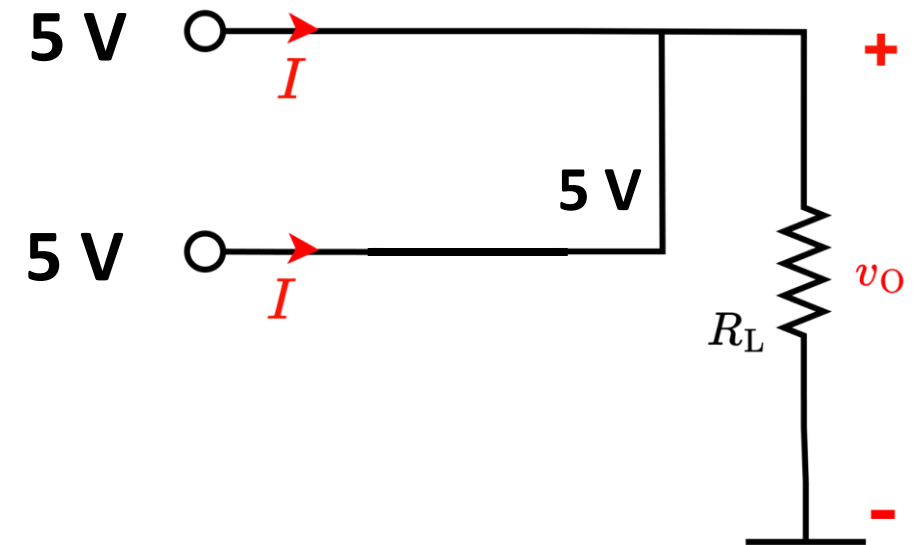
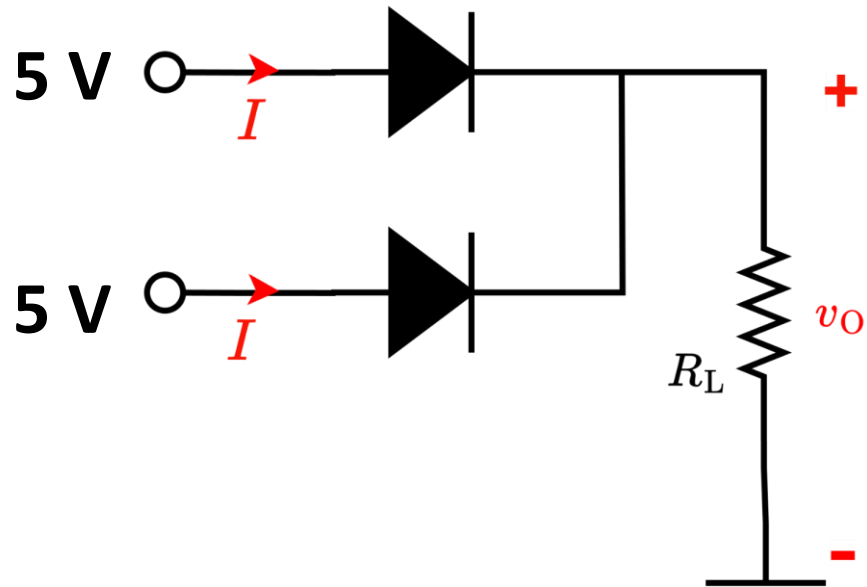


$$Z = 0\text{ V}$$



Logical Operations with Diode (OR)

Ideal diode

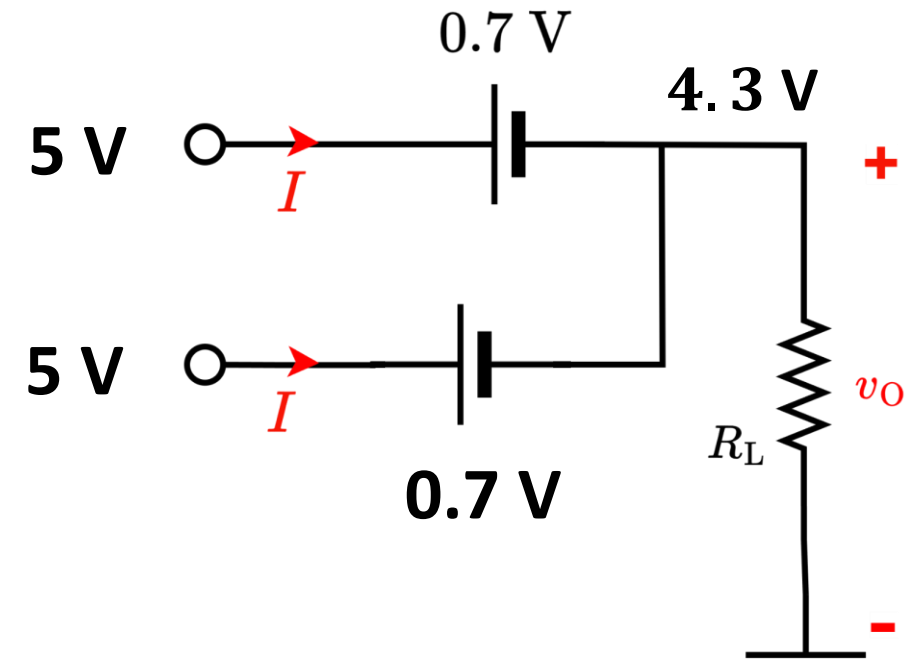
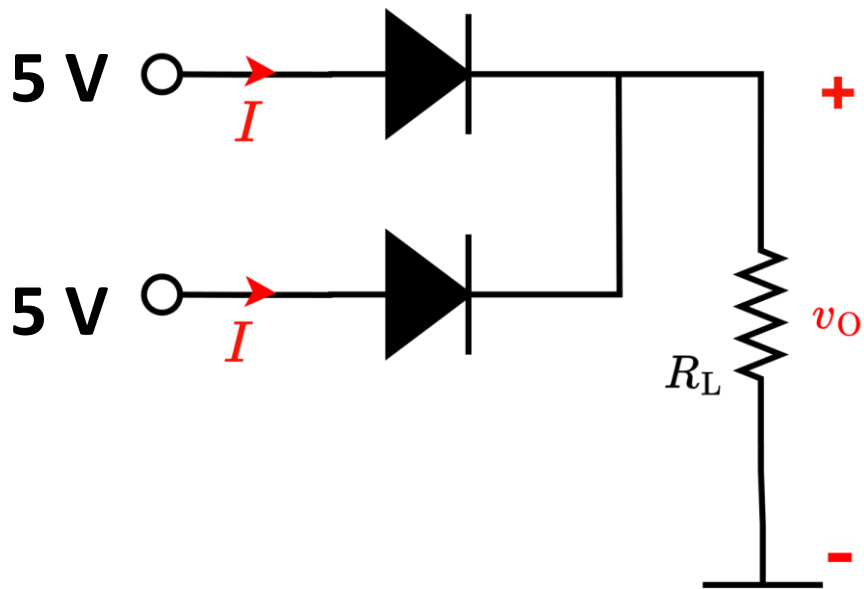


$$Z = 5 \text{ V}$$



Logical Operations with Diode (OR)

CVD diode



$$Z = 4.3 \text{ V}$$

Degraded 5 V

Both diodes have same V_{DO}

Logical Operations with Diode (OR)

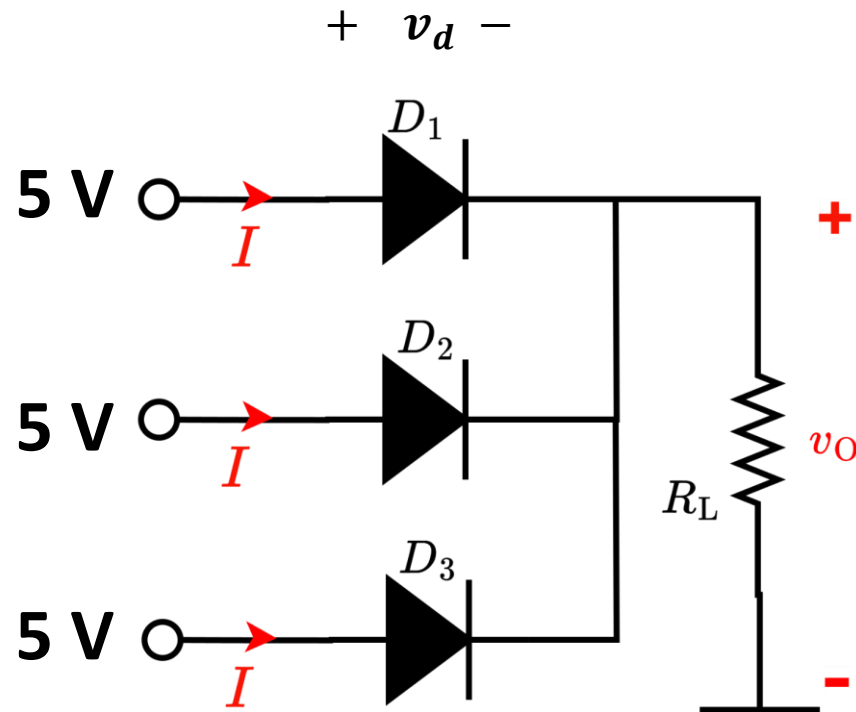
CVD diode

What if the diodes have different V_{D0} ?

$$V_{D1} = 1 \text{ V}$$

$$V_{D2} = 0.7 \text{ V}$$

$$V_{D3} = 0.5 \text{ V}$$



Logical Operations with Diode (OR)

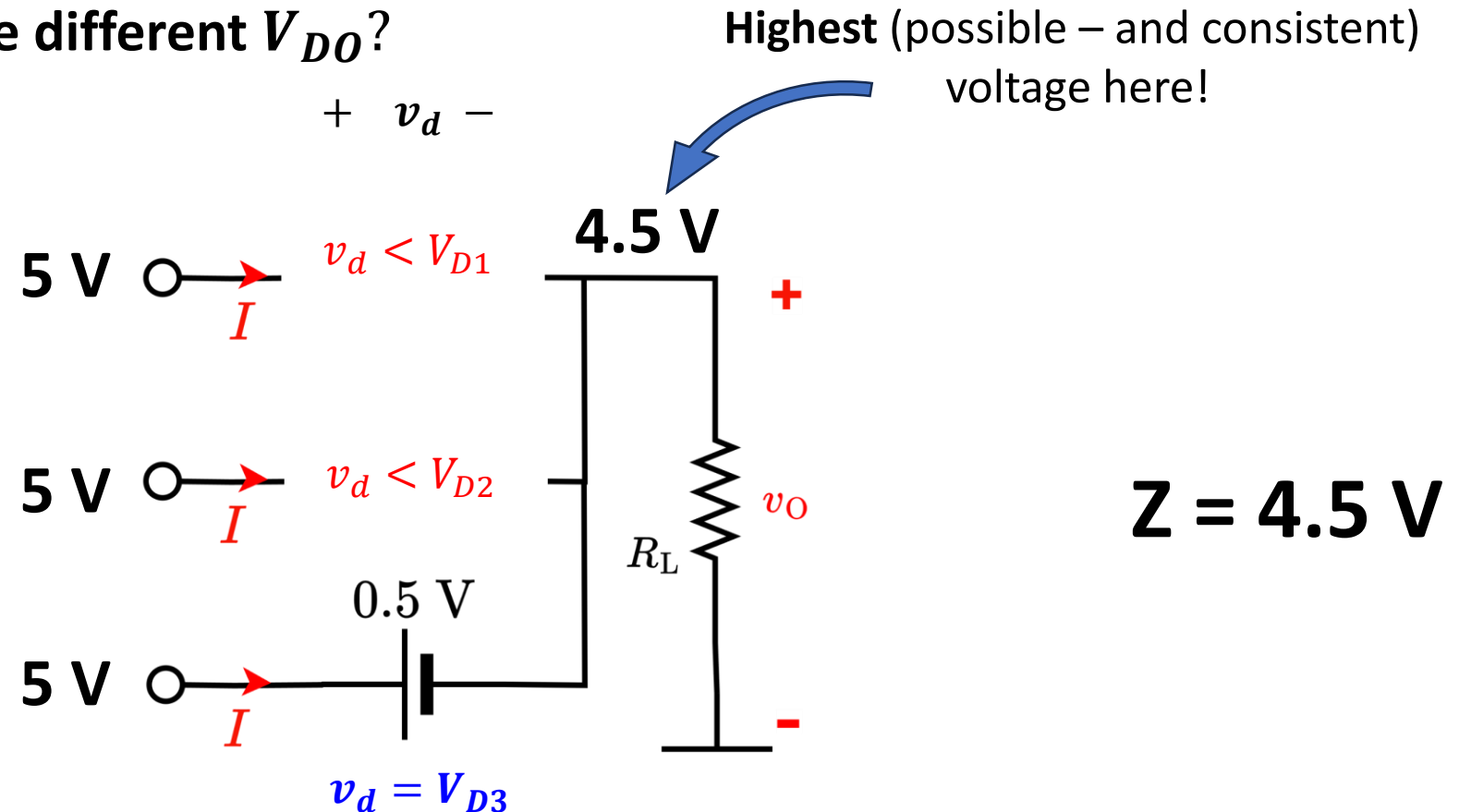
CVD diode

What if two diodes have different V_{D0} ?

$$V_{D1} = 1 \text{ V}$$

$$V_{D2} = 0.7 \text{ V}$$

$$V_{D3} = 0.5 \text{ V}$$



Logical Operations with Diode (OR)

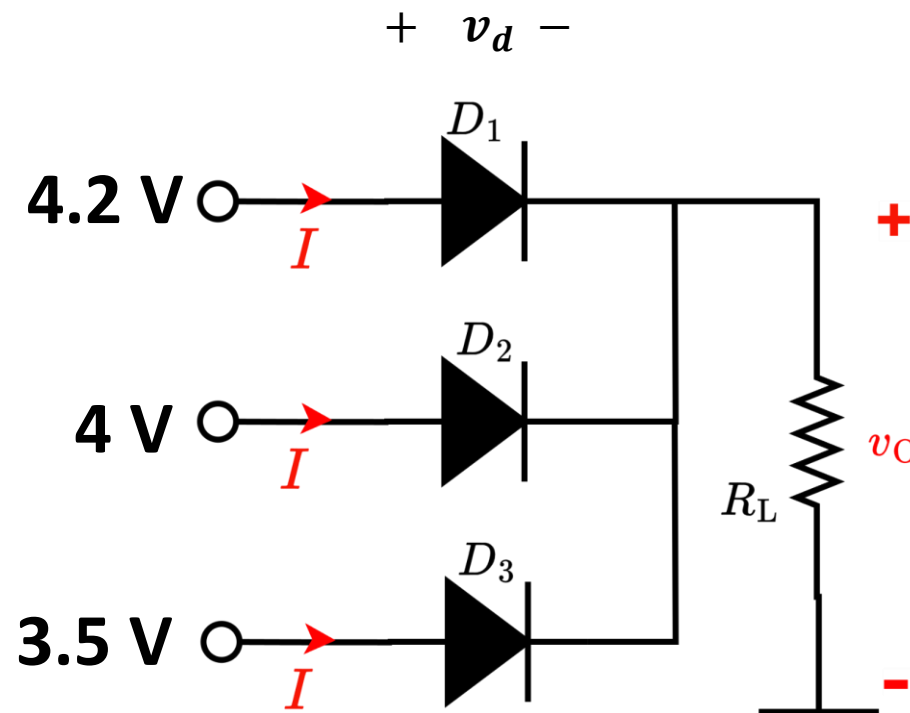
CVD diode

What if the input voltages are different?

$$V_{D1} = 1 \text{ V}$$

$$V_{D2} = 0.7 \text{ V}$$

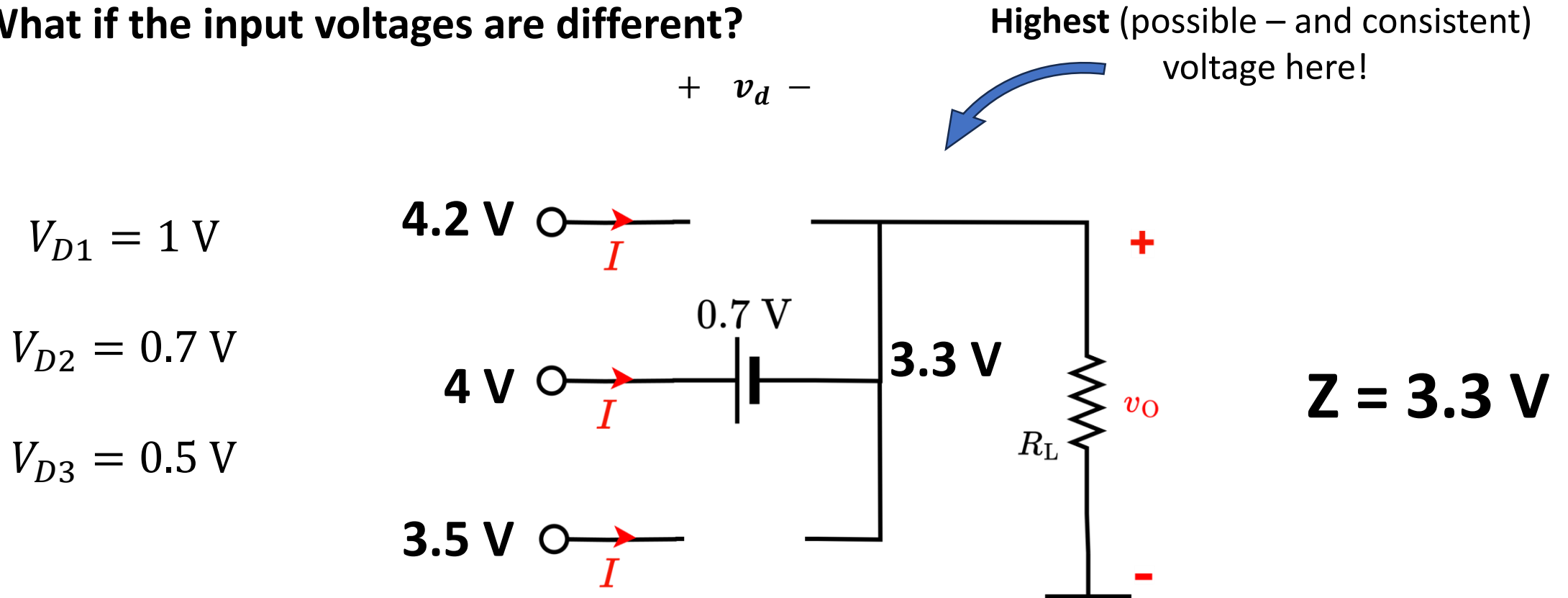
$$V_{D3} = 0.5 \text{ V}$$



Logical Operations with Diode (OR)

CVD diode

What if the input voltages are different?



Logical Operations with Diode (AND)

Logic Truth Table

INPUTS		OUTPUT
X	Y	Z
0	0	0
0	1	0
1	0	0
1	1	1

Logic Levels:

Corresponding voltage levels:

Low/False

0

0V

Voltage Truth Table

INPUTS				OUTPUT
X		Y		Z
0	V	0	V	0 V
0	V	5	V	0 V
5	V	0	V	0 V
5	V	5	V	5 V

High/True

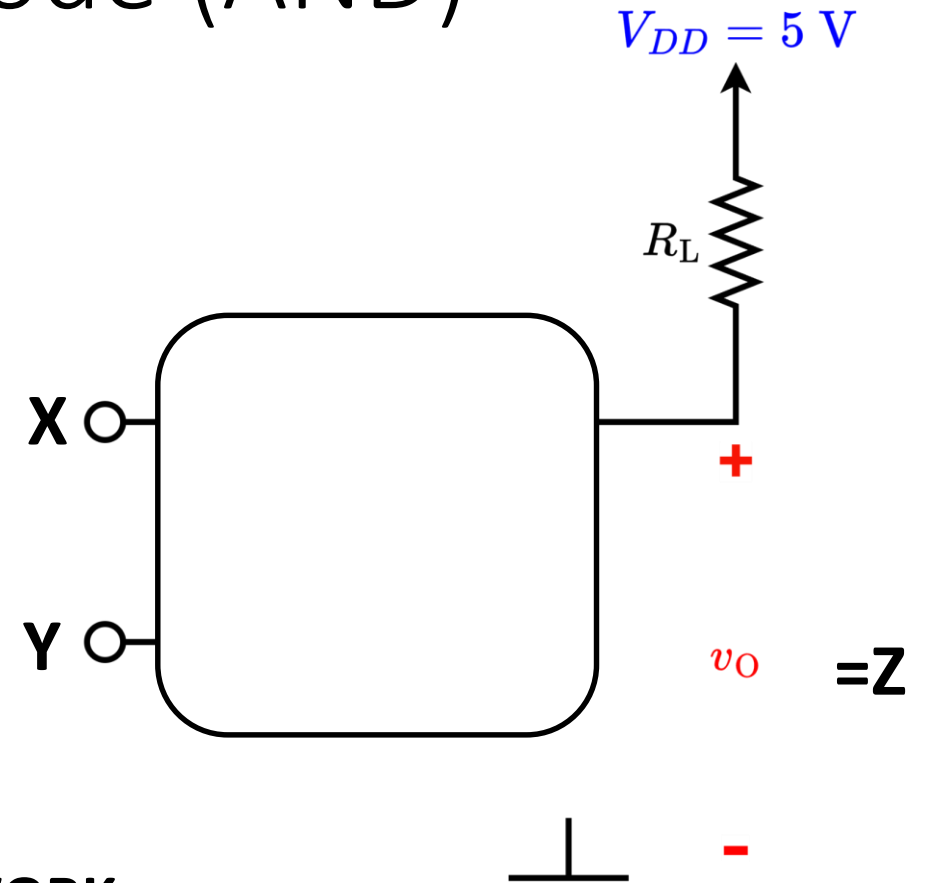
1

5V

Logical Operations with Diode (AND)

Voltage Truth Table

INPUTS		OUTPUT	
X	Y	Z	
0 V	0 V	0 V	
0 V	5 V	0 V	
5 V	0 V	0 V	
5 V	5 V	5 V	



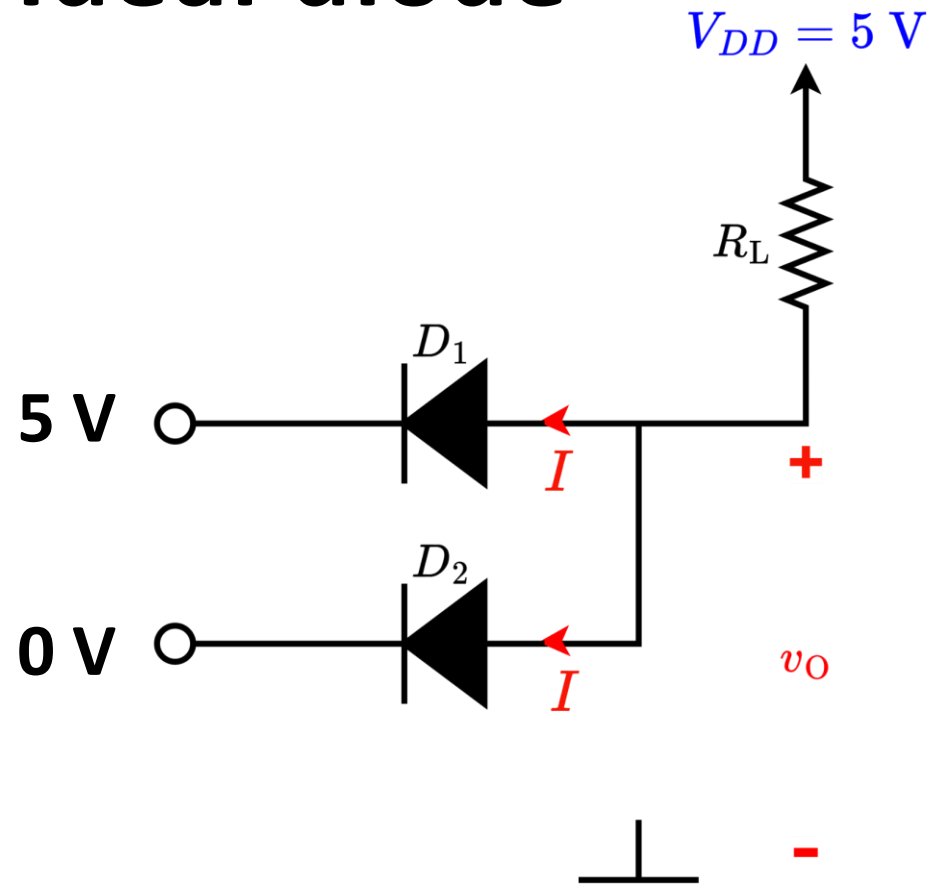
PULL UP NETWORK

When all inputs are completely disconnected, v_O is pulled up to V_{DD}

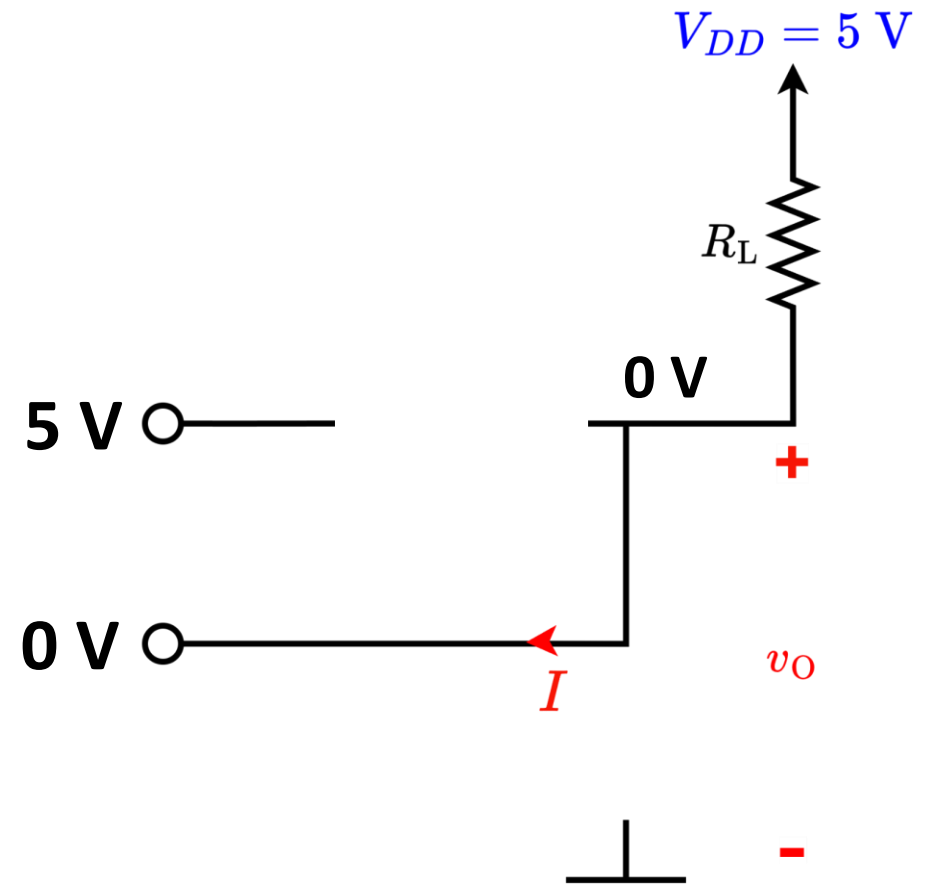
Degrades the LOWEST output voltage

Logical Operations with Diode (AND)

Ideal diode

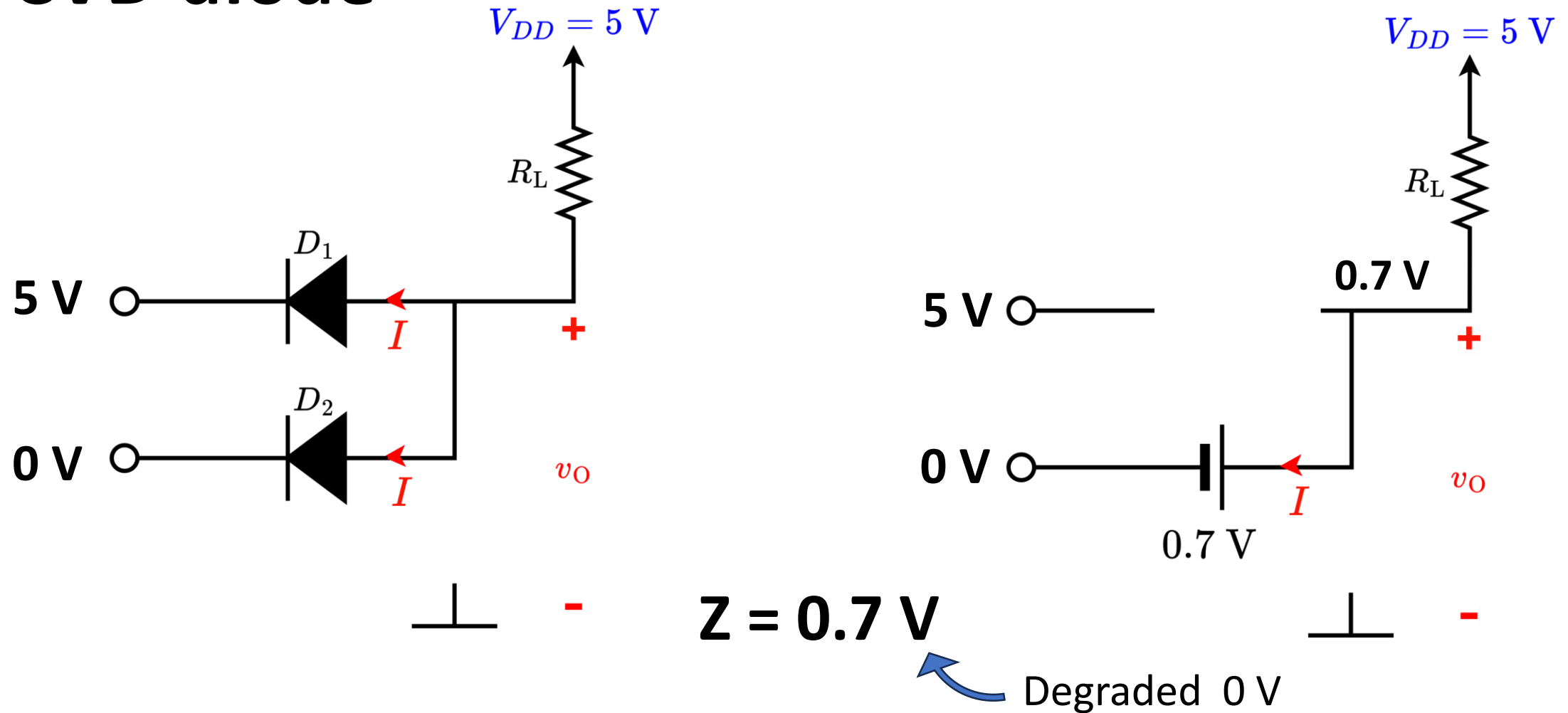


$$Z = 0\text{ V}$$



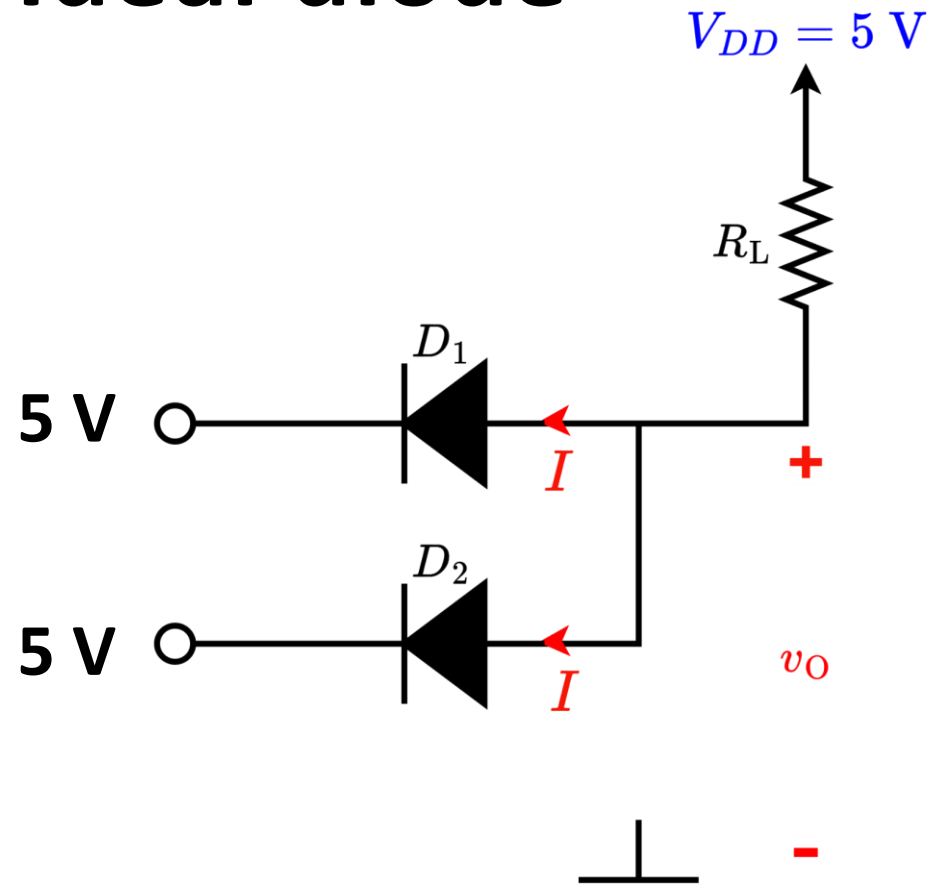
Logical Operations with Diode (AND)

CVD diode

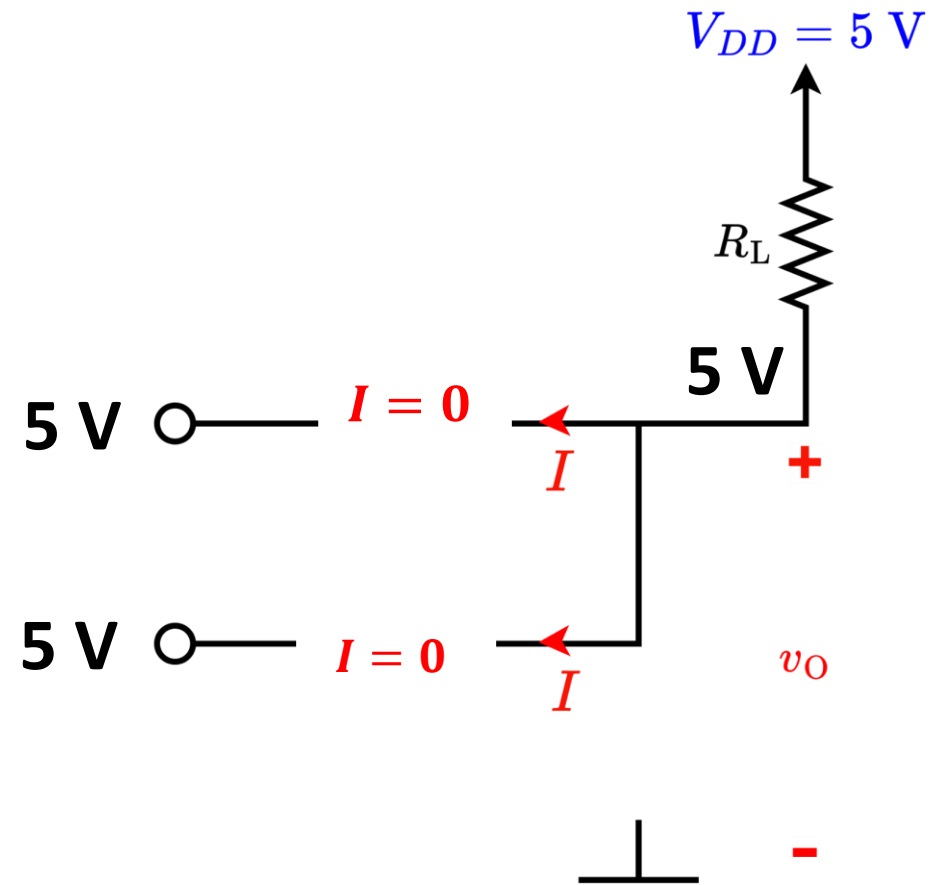


Logical Operations with Diode (AND)

Ideal diode

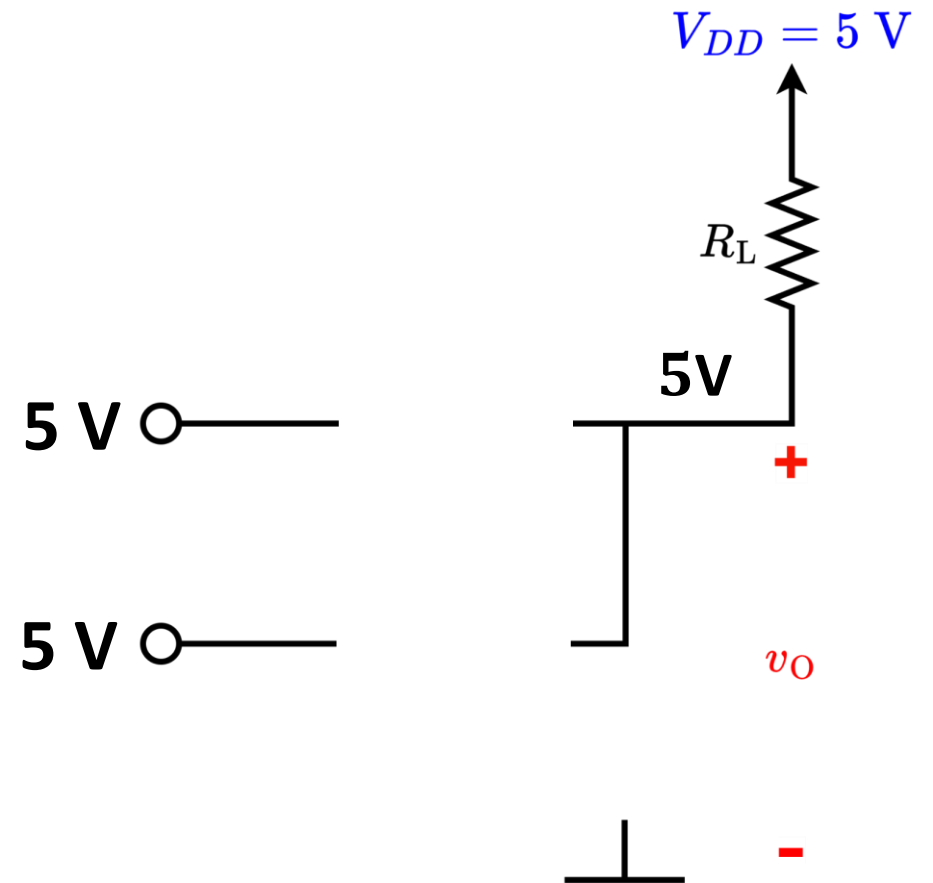
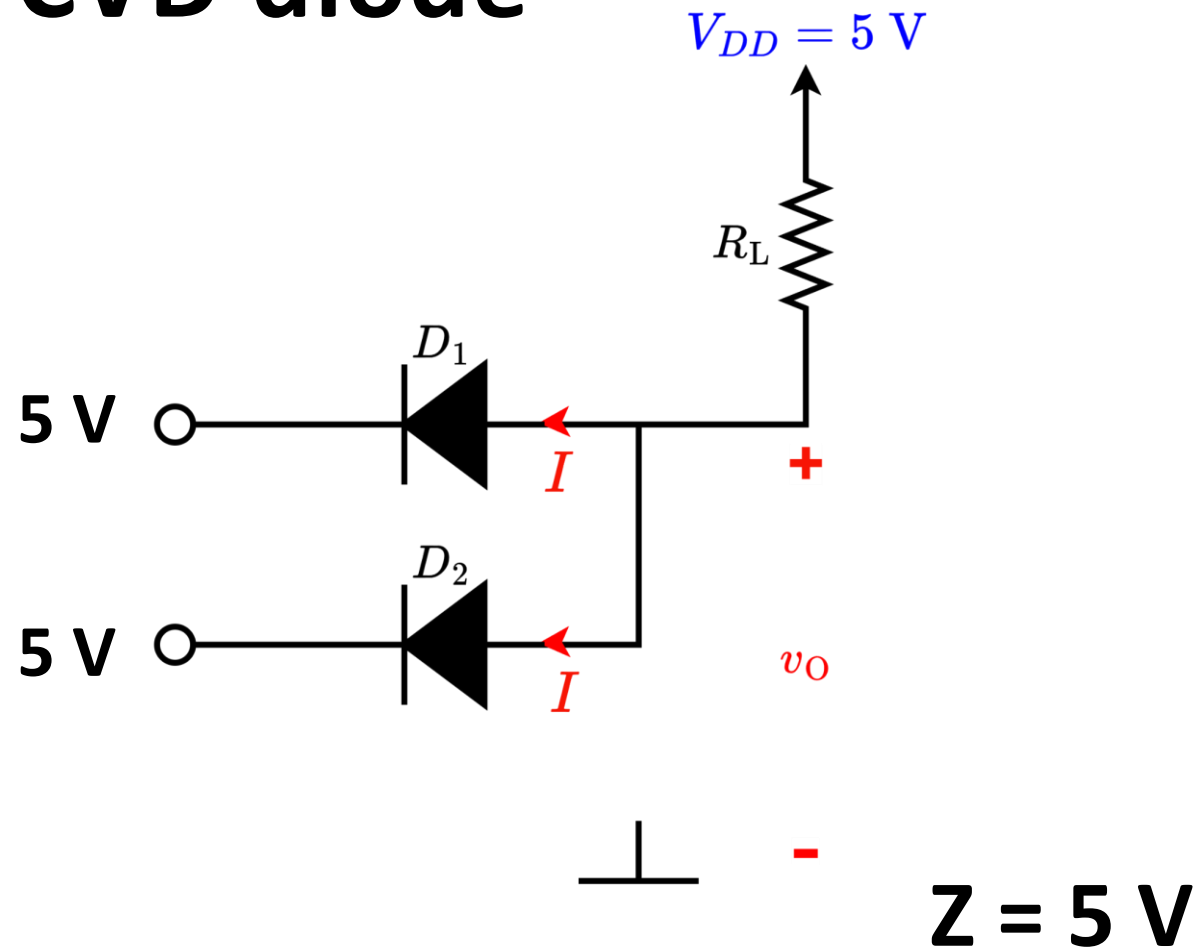


$$Z = 5 \text{ V}$$



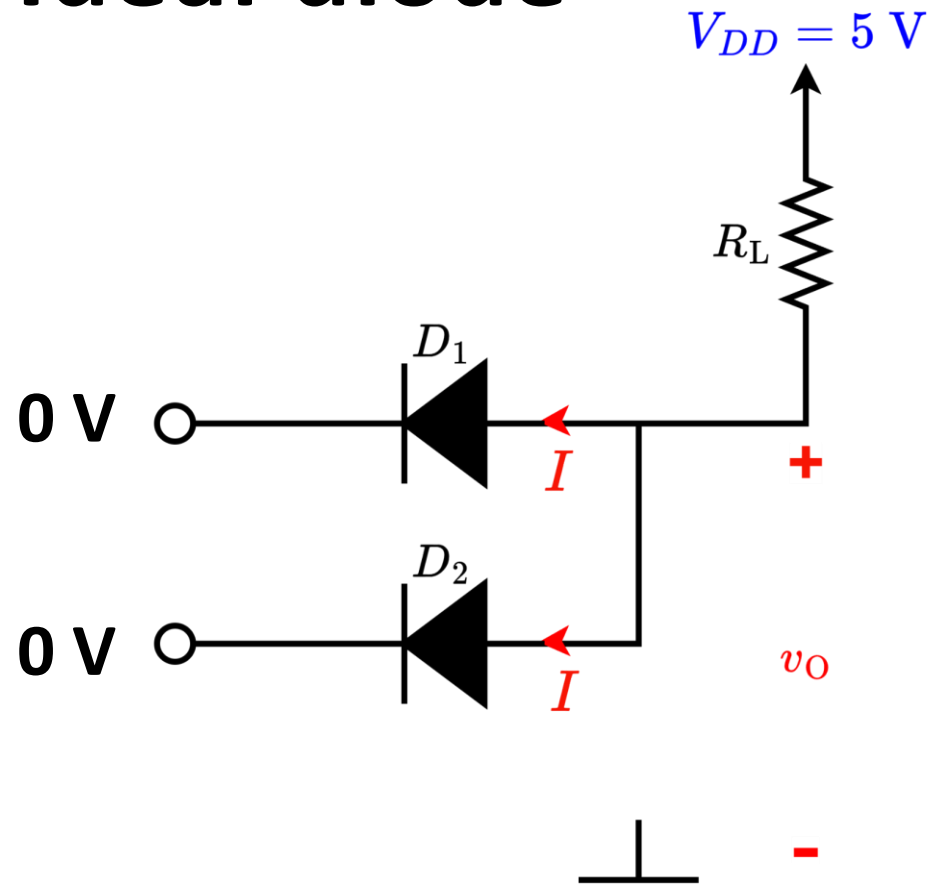
Logical Operations with Diode (AND)

CVD diode

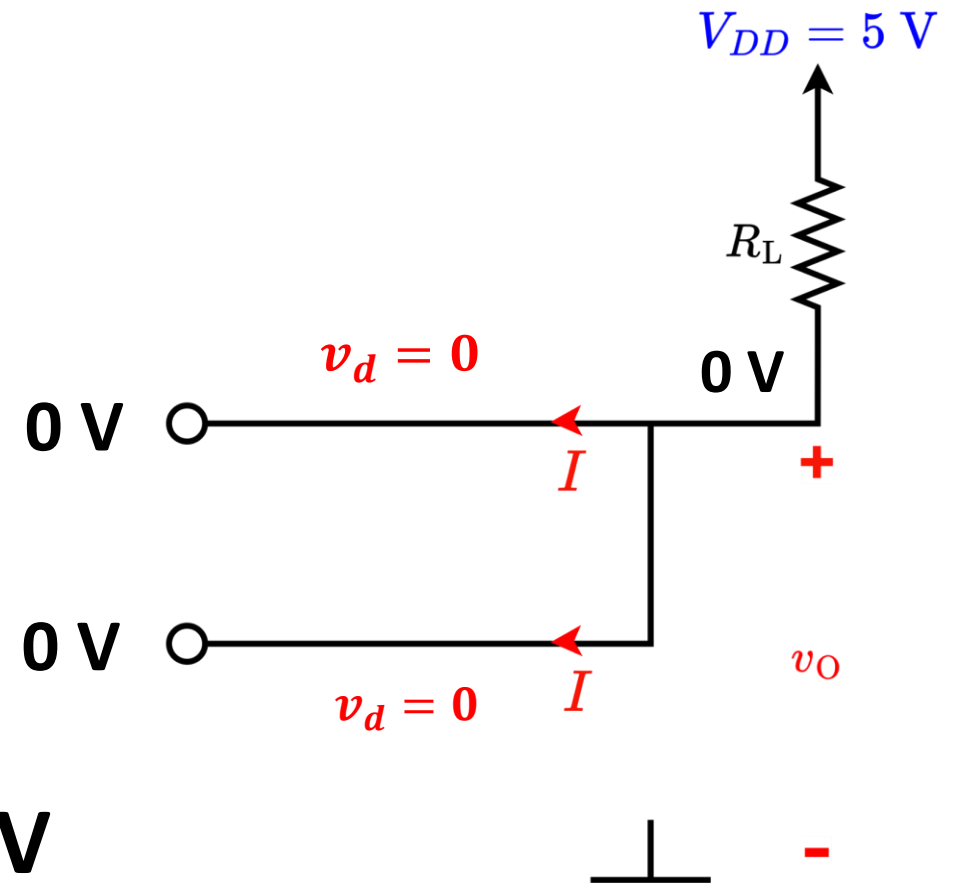


Logical Operations with Diode (AND)

Ideal diode

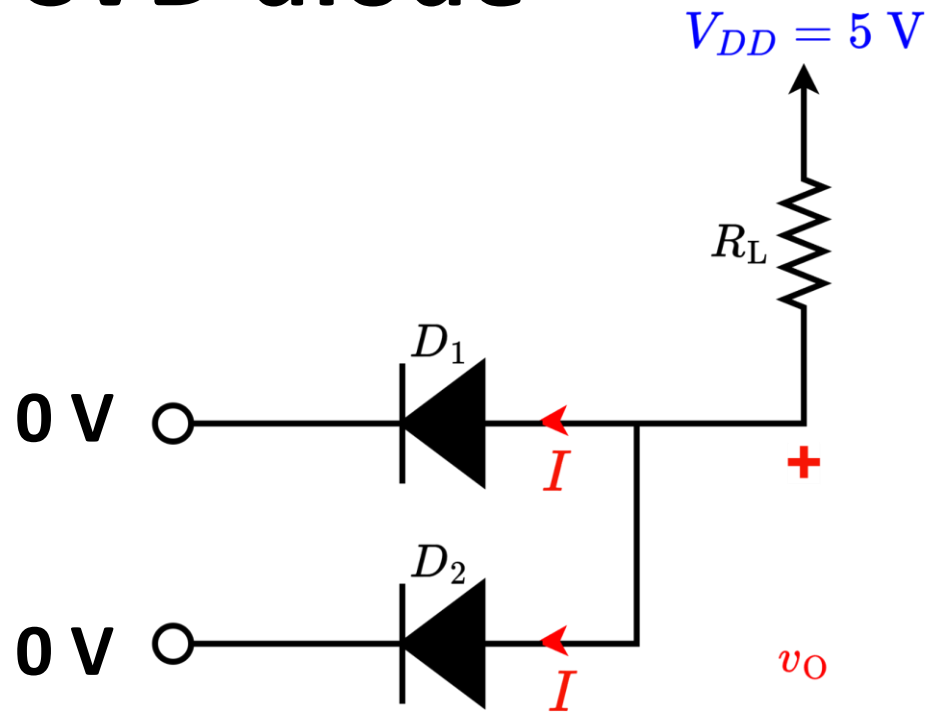


$$Z = 0\text{ V}$$

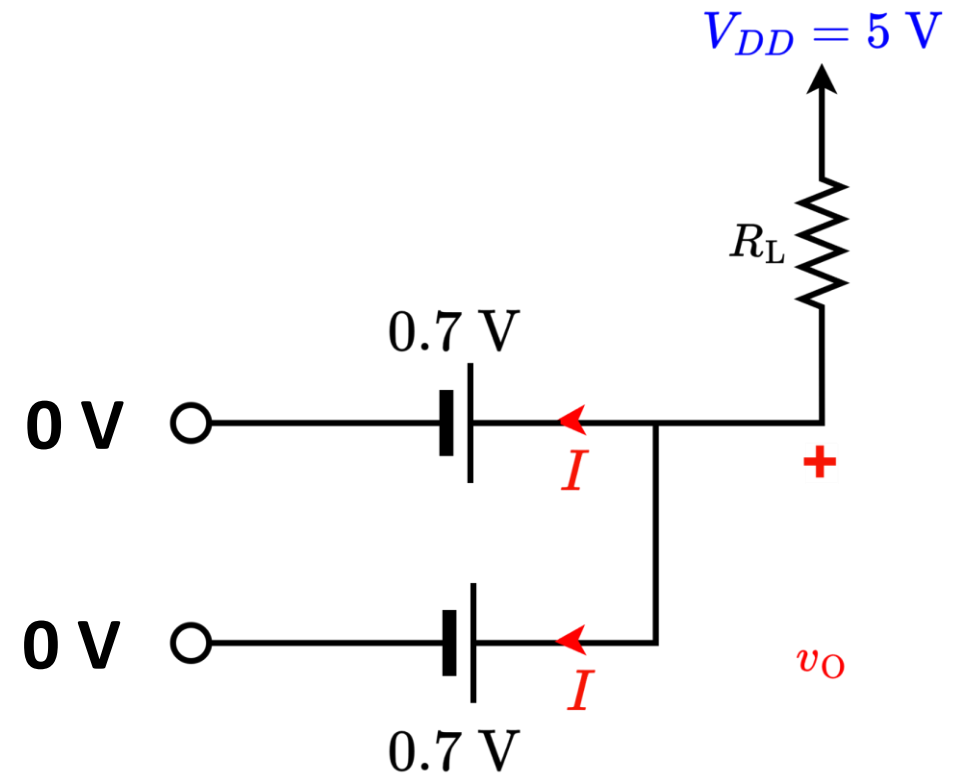


Logical Operations with Diode (AND)

CVD diode



$\perp$ - **$Z = 0.7\text{ V}$**
Degraded 0 V



$\perp$ - Both diodes have same V_{DO}

Logical Operations with Diode (AND)

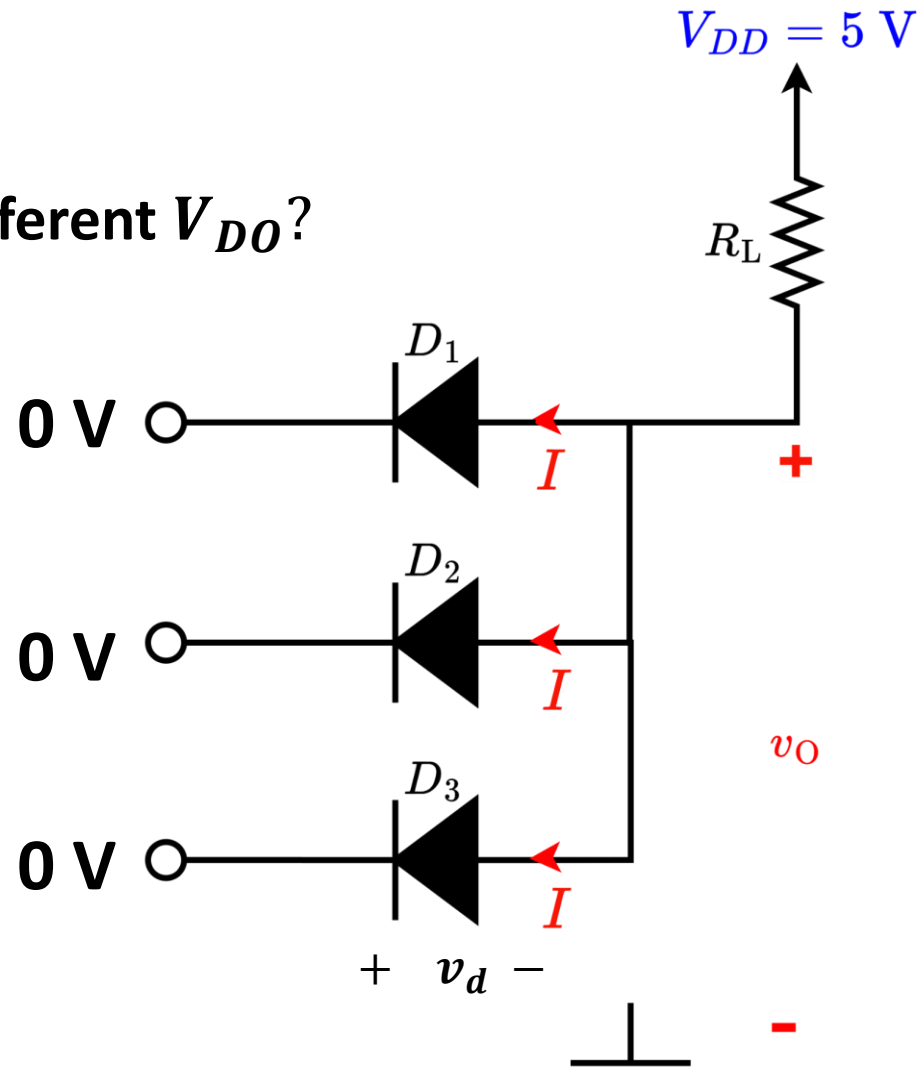
CVD diode

What if the diodes have different V_{DO} ?

$$V_{D1} = 1 \text{ V}$$

$$V_{D2} = 0.7 \text{ V}$$

$$V_{D3} = 0.5 \text{ V}$$



Logical Operations with Diode (AND)

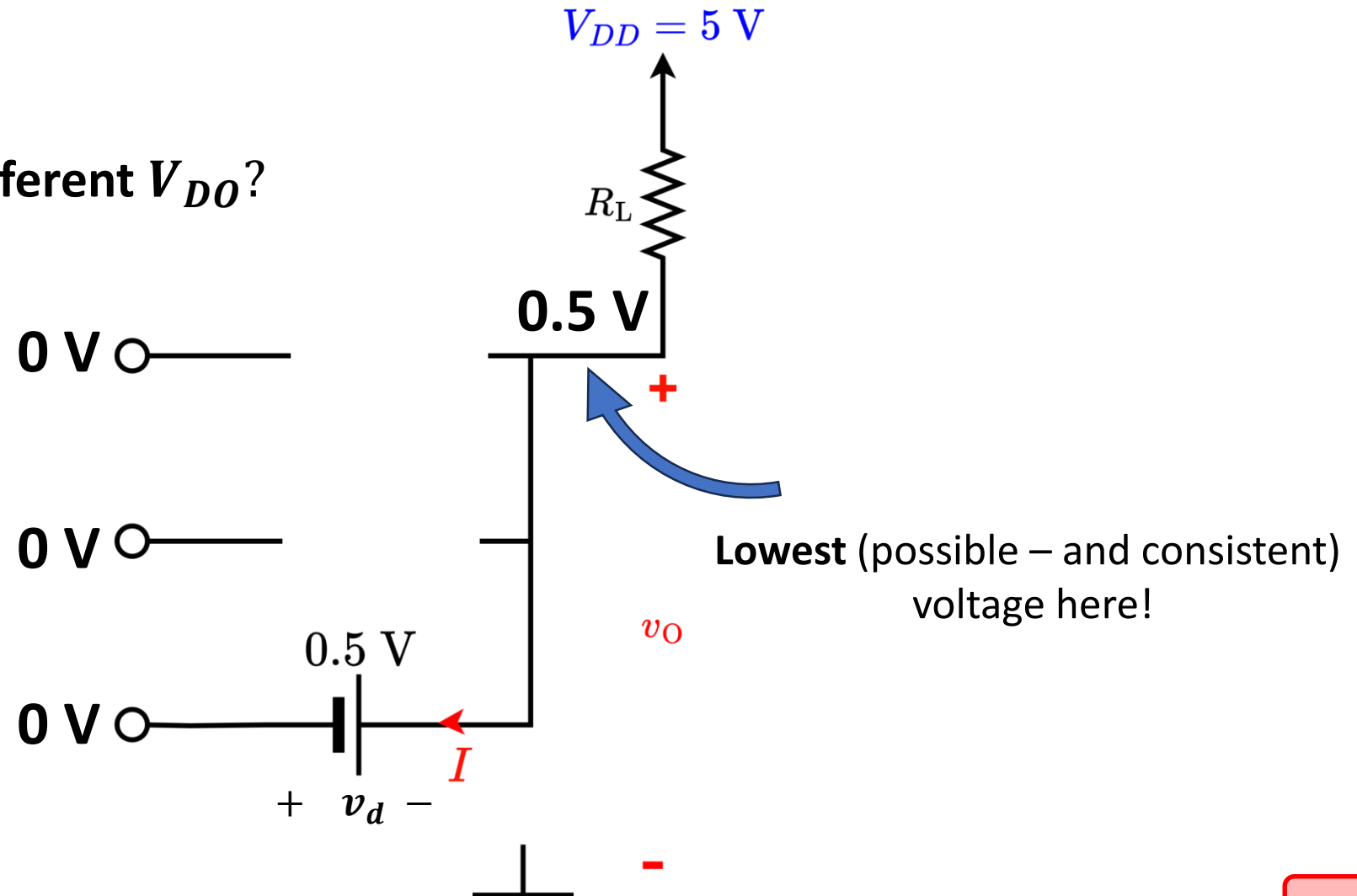
CVD diode

What if the diodes have different V_{DO} ?

$$V_{D1} = 1 \text{ V}$$

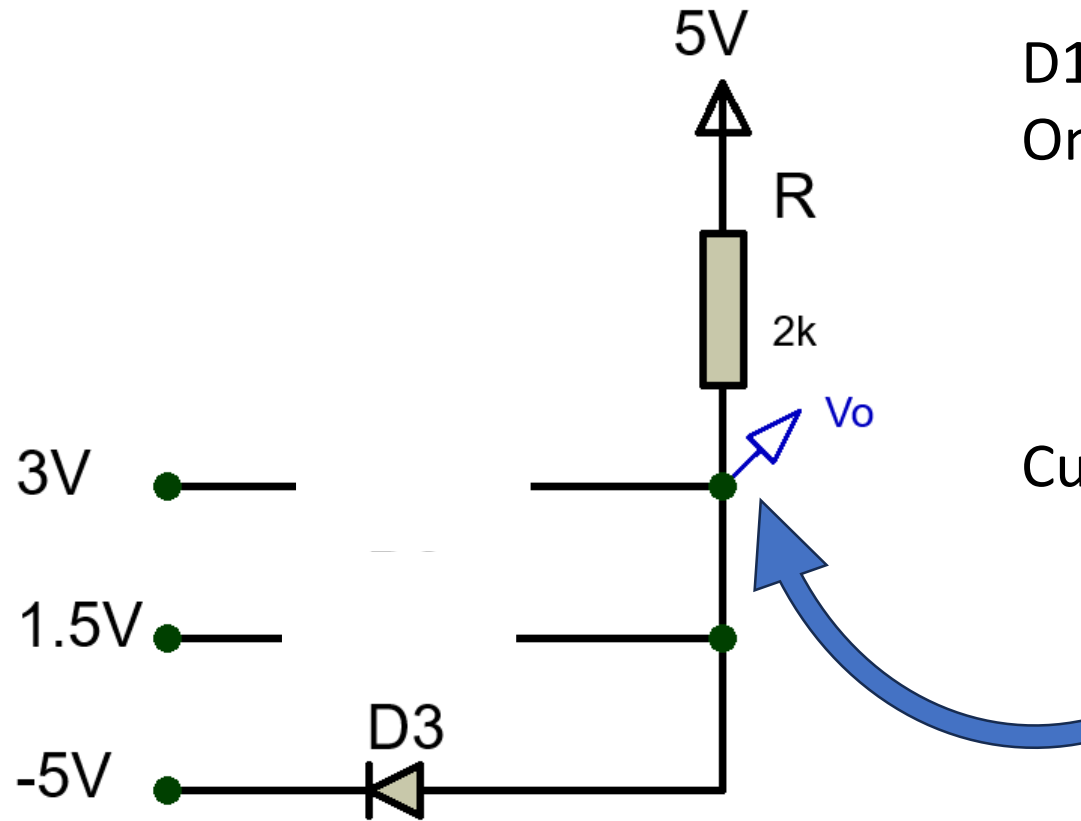
$$V_{D2} = 0.7 \text{ V}$$

$$V_{D3} = 0.5 \text{ V}$$



Effect of input Voltage Variation in Logic Gates (AND)

Example 1: Find the value of V_o (All the diodes are ideal diodes)



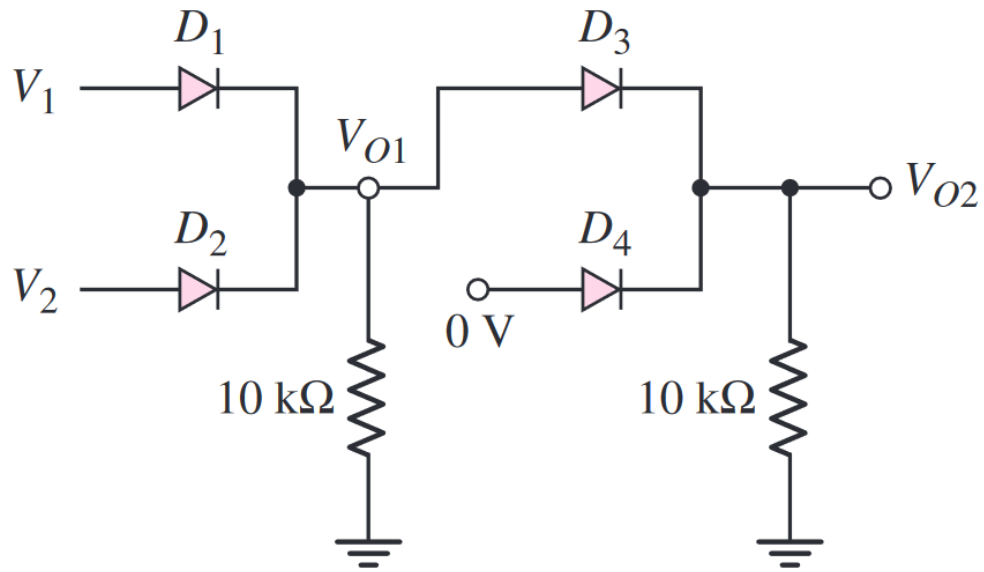
D1 and D2 are OFF $\rightarrow$ Open Circuit
Only D3 is ON $\rightarrow$ Short Circuit

$$V_o = -5 \text{ V}$$

$$\text{Current, } I = \frac{5 \text{ V} - (-5 \text{ V})}{2 \text{ k}\Omega} = 5 \text{ mA}$$

Lowest (possible – and consistent)
voltage here!

Combined Logic Circuits

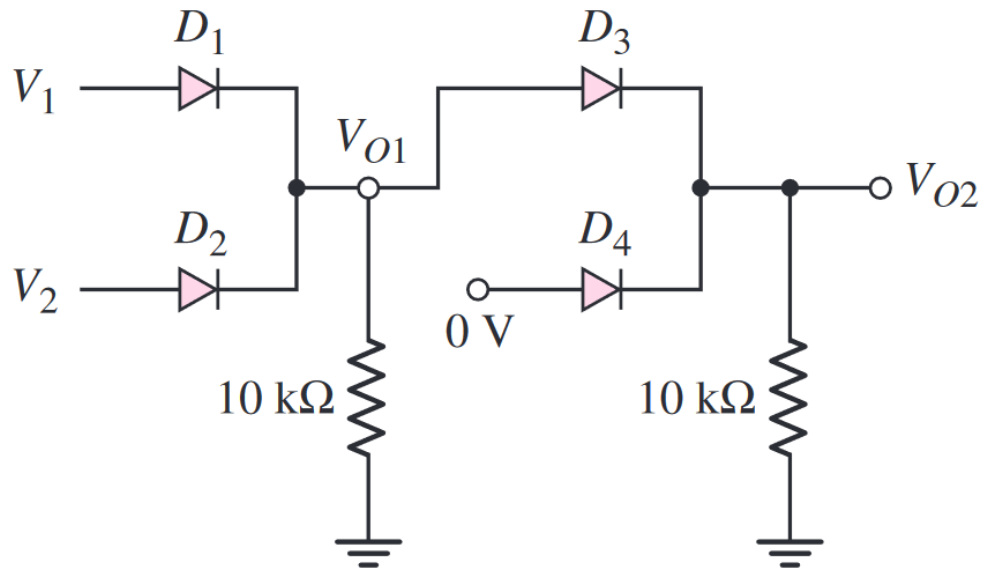


$$V_{O1} = V_1 \text{ OR } V_2 = V_1 | V_2$$

$$V_{O2} = (V_{O1} \text{ OR } 0) = V_{O1} = V_1 | V_2$$

Combined Logic Circuits

Example 2:



** In CVD diode models, we are assuming that all diodes have equal drop.

Suppose: $V_1 = 3\text{ V}$, $V_2 = 2\text{ V}$.

For Ideal diodes assumption:

$$V_{O1} = V_1 \text{ OR } V_2 \rightarrow \text{Largest Value of the inputs}$$

$$\therefore V_{O1} = 3\text{ V}$$

$$V_{O2} = (V_{O1} \text{ OR } 0) = V_{O1} = 3\text{ V}$$

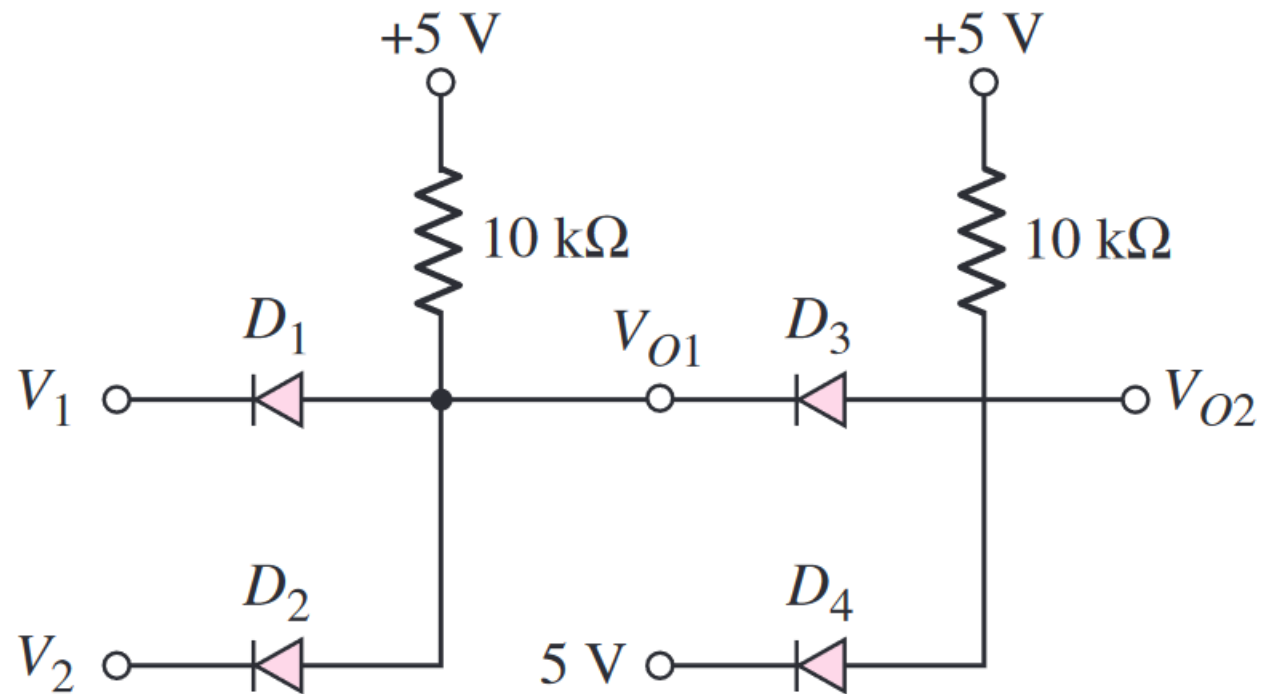
For CVD diodes assumption:

$$V_{O1} = V_1 \text{ OR } V_2 \rightarrow \text{Largest Value of the inputs} - V_{DO}$$

$$\therefore V_{O1} = (3 - V_{DO})\text{ V}$$

$$V_{O2} = (V_{O1} \text{ OR } 0) = V_{O1} - V_{DO} = (3 - 2 V_{DO})\text{ V}$$

Combined Logic Circuits

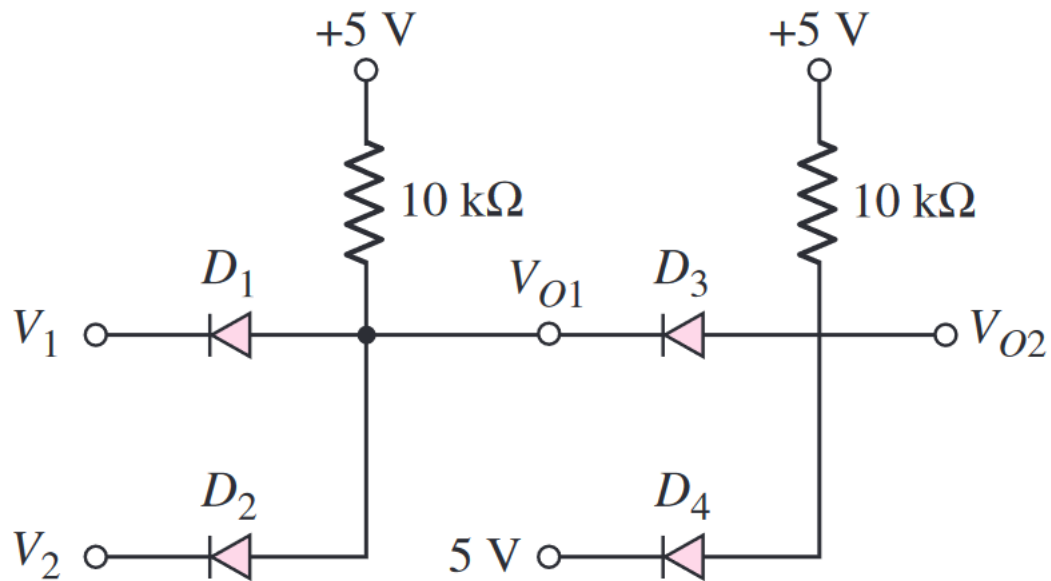


$$V_{O1} = V_1 \text{ AND } V_2 = V_1 \cdot V_2$$

$$V_{O2} = (V_{O1} \text{ AND } 5) = V_{O1} = V_1 \cdot V_2$$

Combined Logic Circuits

Example 3:



** In CVD diode models, we are assuming that all diodes have equal drop.

Suppose: $V_1 = 3\text{ V}$, $V_2 = 1.5\text{ V}$.

For Ideal diodes assumption:

$V_{O1} = V_1 \text{ AND } V_2 \rightarrow \text{Smallest Value of the inputs}$

$$\therefore V_{O1} = 1.5\text{ V}$$

$$V_{O2} = (V_{O1} \text{ AND } 5) = V_{O1} = \mathbf{1.5\text{ V}}$$

For CVD diodes assumption:

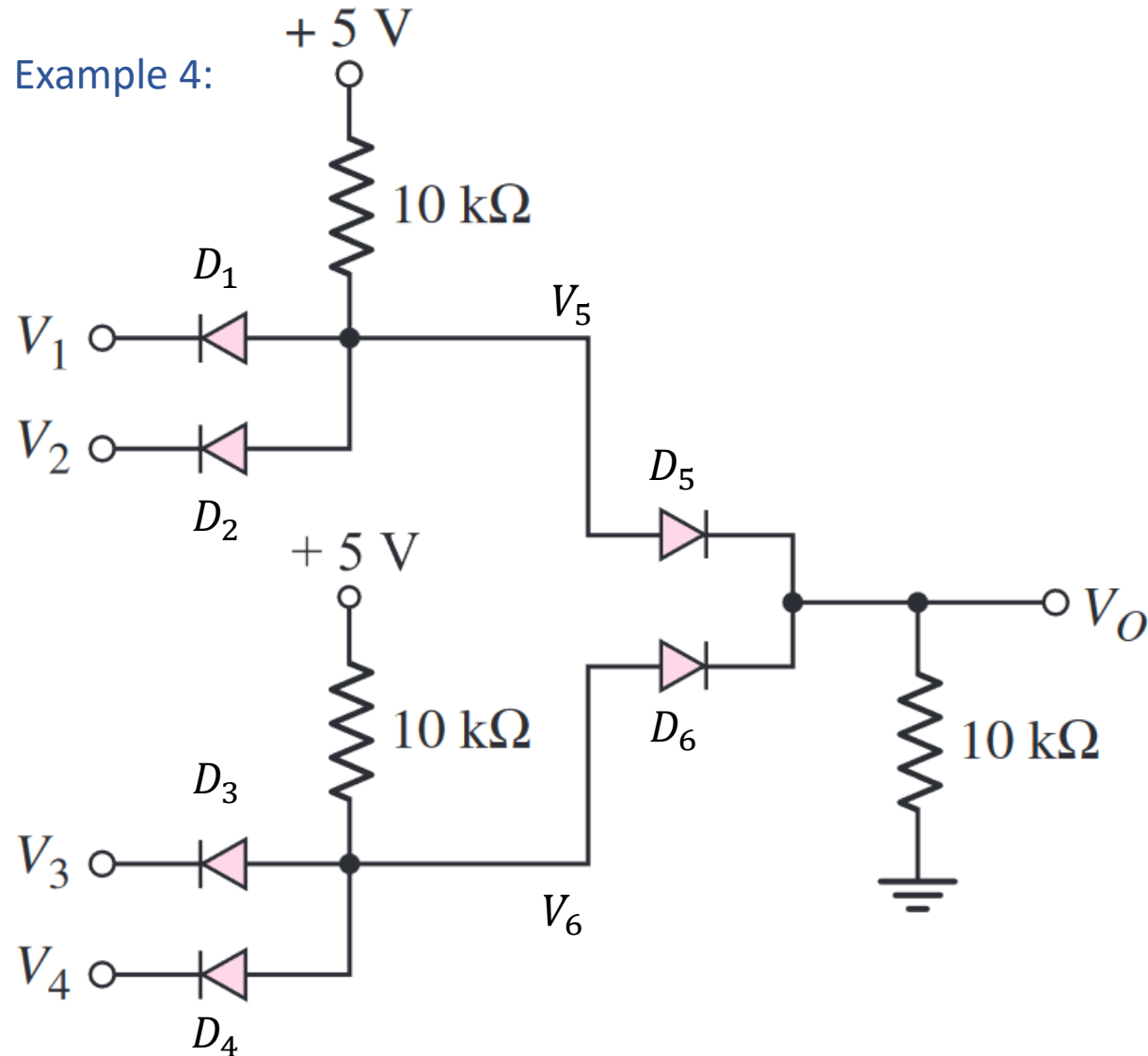
$V_{O1} = V_1 \text{ AND } V_2 \rightarrow \text{Smallest Value of the inputs} + V_{DO}$

$$\therefore V_{O1} = (1.5 + V_{DO})\text{ V}$$

$$V_{O2} = (V_{O1} \text{ AND } 5) = V_{O1} + V_{DO} = (\mathbf{1.5 + 2 V_{DO}})\text{ V}$$

Combined Logic Circuits

Example 4:



Express V_0 as a Boolean expression of V_1, V_2, V_3 and V_4

$$V_0 = (V_1 \cdot V_2) | (V_3 \cdot V_4)$$

$$V_0 = (V_1 \text{ AND } V_2) \text{ OR } (V_3 \text{ AND } V_4)$$

Thank You!